

No. of Functional Features Included in the Solution

Date	29 September 2026
Team ID	6ab4ce760fc666a751b52130
Project Name	AI StudyBuddy
Maximum Marks	5 Marks

Functional Features Overview

S.No	Feature Name	Feature Description	Module / Component	Status	Marks
1	User Registration	Register new user with name, email, password. Password hashed with bcrypt. Returns JWT tokens.	authController.js	Done	Core
2	User Login	Authenticate user with email and password. Returns JWT access and refresh tokens.	authController.js	Done	Core
3	Token Refresh	Issue new JWT token pair using valid refresh token without re-login.	authController.js	Done	Core
4	User Logout	Stateless logout — server acknowledges, client discards tokens.	authController.js	Done	Supporting
5	Upload Study Material	Upload .txt, .md, .pdf files (max 5MB). Text content extracted and stored in MongoDB. Temp file deleted.	materialController.js, upload.js	Done	Core
6	List All Materials	Return all materials belonging to the authenticated user. Admins see all. Heavy fields excluded for speed.	materialController.js	Done	Core
7	Get Single Material	Return full material details including content and all AI-generated data. Ownership check enforced.	materialController.js	Done	Supporting
8	Delete Material	Delete a material. Students can only delete own materials. Admins can delete any.	materialController.js	Done	Supporting
9	AI Summary Generation	Generate concise bullet-point summary from material content using Google Gemini AI. Saved to DB.	materialController.js, gemini.js	Done	Core
10	AI Flashcard Generation	Generate Q&A flashcards (JSON format) from material. Count configurable (default: 5). Saved to DB.	materialController.js, aiController.js, gemini.js	Done	Core
11	AI MCQ Quiz Generation	Generate MCQ questions with 4 options and correct answer (JSON). Count configurable. Saved to DB.	materialController.js, aiController.js, gemini.js	Done	Core
12	AI Study Plan Generation	Generate personalized day-by-day study schedule based on goal, hoursPerDay, and days. Saved to DB.	materialController.js, aiController.js, gemini.js	Done	Core
13	Admin — List All Users	View all registered users (passwords excluded). Admin role required.	adminController.js	Done	Supporting
14	Admin — Delete User	Delete a user and all their associated materials. Admin role required.	adminController.js	Done	Supporting
15	Admin — Platform Statistics	View total user count and total materials count. Admin role required.	adminController.js	Done	Supporting
16	JWT Authentication Middleware	Verify Bearer token on all protected routes. Extract userId and role from token.	middleware/auth.js	Done	Core
17	Role-Based Access Control	adminOnly middleware restricts admin routes. Ownership checks in controllers.	middleware/auth.js, controllers	Done	Core
18	Health Check	GET / endpoint returns server status message for monitoring.	index.js	Done	Supporting

Feature Summary

Metric	Count / Value
Total Features Planned	18
Total Features Implemented	18
Core / Must-Have Features	11
Additional / Nice-to-Have Features	7
Features Tested & Verified	18

Feature Category Breakdown

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	0 (Backend-only API; Postman used as client)	N/A — REST API project
2	Backend / Logic	12	Register, Login, Token Refresh, Logout, Upload Material, List Materials, Get Material, Delete Material, AI Summarize, AI Flashcards, AI Quiz, AI Study Plan
3	Database / Storage	2	MongoDB Users Collection, MongoDB Materials Collection (with embedded AI content)
4	API / Integration	2	Google Gemini 2.5 Flash API integration, 18 REST API endpoints
5	Security / Authentication	2	JWT Authentication Middleware (protect), Role-Based Access Control (adminOnly)